PHILIPPINE ARMY MANUAL 6-0102-04

CYBERSPACE OPERATIONS

Proponent: Office of the Assistant Chief of Staff for Command and Control Communications, and Cyber System, G6, Philippine Army
Promulgated: 24 February 2020

PHILIPPINE ARMY MANUAL 6-0102-04



CYBERSPACE OPERATIONS

Proponent: Office of the Assistant Chief of Staff for Command and Control Communications, and Cyber System, G6, Philippine Army
Promulgated: 24 February 2020

PHILIPPINE ARMY MANUAL 6-0102-04

CYBERSPACE OPERATIONS

Philippine Army Manual 6-0102-04 (PAM 6-0102-04) is promulgated
on 24 February 2020 by authority of the Commanding General, Philippine Army



LTGEN GILBERT I GAPAY AFP
Commanding General, Philippine Army

Preface

1. The passage of GHQ Letter Directive 28 recognizing cyberspace as one of the domains of Armed Forces of the Philippines operation, brought a question on how the Philippine Army (PA) will operate in this new type of domain. While the Army is not new to emerging information technologies, the challenge is how these technologies could be doctrinally part of an operation or be an operation itself.

2. This manual sets forth the principal concepts, guidelines and other considerations in conducting Cyberspace Operations. It puts into context both its similarity as well as distinction against regular combat operations, especially in terms of planning and execution. More importantly, this manual shall identify how cyberspace operations fits in to the overall operational concept of the Philippine Army and with other Major Services in joint operations.

3. This manual also serves as reference to the following:

- Army commanders who wish to take full advantage of the available cyber assets in support of their mission
- Signal officers and personnel who shall be at forefront of delivering cyber related services to their supported units
- Sustainment planners tasked to sustain this type of operation.
- Other major service 6 for their consideration in conducting joint operation with the Philippine Army.

4. The proponent of this manual is the Office of the Assistant Chief of Staff for Command and Control, Communications and Cyber Systems, C4S, G6, PA. However the Doctrine and Capability Integration Center, TRADOC, PA is responsible in the review and updating of this publication. Send comments and recommendations to the Commander, Training and Doctrine Command, Philippine Army, Camp O'Donnell, Sta. Lucia, Capas, Tarlac, (Attn: Director, DACIC).

5. Policies, rules, regulations and other publications inconsistent to this publication are hereby rescinded.

6. Unless this publication states otherwise, masculine nouns and pronouns do not refer exclusively to men.

Contents **Page**

Chapter I – Philippine Army Operations in Cyberspace 1
The Cyberspace 1
Cyberspace Operational Environment 1
Threat Actors 2
PA Cyberspace Operations Mission 3
PA Cyberspace Operational Framework 4
Chapter II – Fundamentals of Cyberspace Operations. 5
Concepts and Strategies 5
Cyberspace Operations Activities 6
Roles and Responsibilities 8
Chapter III – Planning and Execution 10
State of the Operational Environment 10
Current Threats 11
End State 11
Phases of Cyberspace Operations 11
Cyberspace Operational Design 11
PA Cyberspace Operations by Echelon 12
PA Cyberspace Operations in Joint Operations 13
PA Cyberspace Operations and the PA C4ISTAR. 13
Chapter IV – Sustaining Cyberspace Operations 14
Personnel Management 14
Cyberspace Operations Mission Essential Equipment 15
Equipment Life Sustainment Program 16
Glossary 17
Abbreviation and Acronym List 22
References 24
Figure list
Figure 1.1 – Domains of Cyberspace Operational Environment (US Joint Chief of Staff, 2018) 1
Figure 1.2 – Cyberspace Operational Framework 4
Figure 2.1 – Operating Environment, Philippine Army Operating Concept (Philippine Army, 2015) 6
Figure 2.2 – Strategic Security Landscape, LMC (PA, n.d.) 7
Figure 3.1 – Cyberspace Operational Design 12
Figure 3.2 – PA Cyberspace Operations by Echelon 12
Figure 4.1 – Laddered Career and Skills Diagram for Cyber Workforce 15
Table list
Table 1.1 – Cyberspace Operations Mission Objectives and Activities 3
Table 3.1 – CSIRT Capabilities by Echelon 13
Table 4.1 – Subdivision of Signal Corps AFPOS (AC of S for C4S, G6, 2017) . 14

Chapter I
Philippine Army Operations in Cyberspace

0101. The Armed Forces of the Philippines (AFP), recognizing cyberspace as one of the domains of military operation, brought a significant paradigm shift in the Philippine Army's (PA) operational environment (OE). Cyberspace, being a borderless domain, transcends the PA traditional territorial-oriented OE. To address the challenges of cyberspace, formulation of a new operational concept is in order. But before any concept can be formulated, the Army needs to have a clear understanding of the meaning of cyberspace and cyberspace operations.

The Cyberspace

0102. Cyberspace is a global domain within the information environment consisting of the interdependent network of information technology infrastructure and residing data, including the internet, telecom networks, computer systems and embedded processors and controllers (Theohary, 2018). The technologies described in the information environment are collectively called cyberspace capabilities or cyber assets.

0103. Cyberspace operations (CO) is the employment of cyberspace capabilities, where the primary purpose is to achieve objectives in or through cyberspace (US Joint Chief of Staff, 2018). Through the years, the use of cyberspace capabilities has become more and more prevalent, and like any military outfit, the PA seeks to use these capabilities to its operational advantage.

Cyberspace Operational Environment

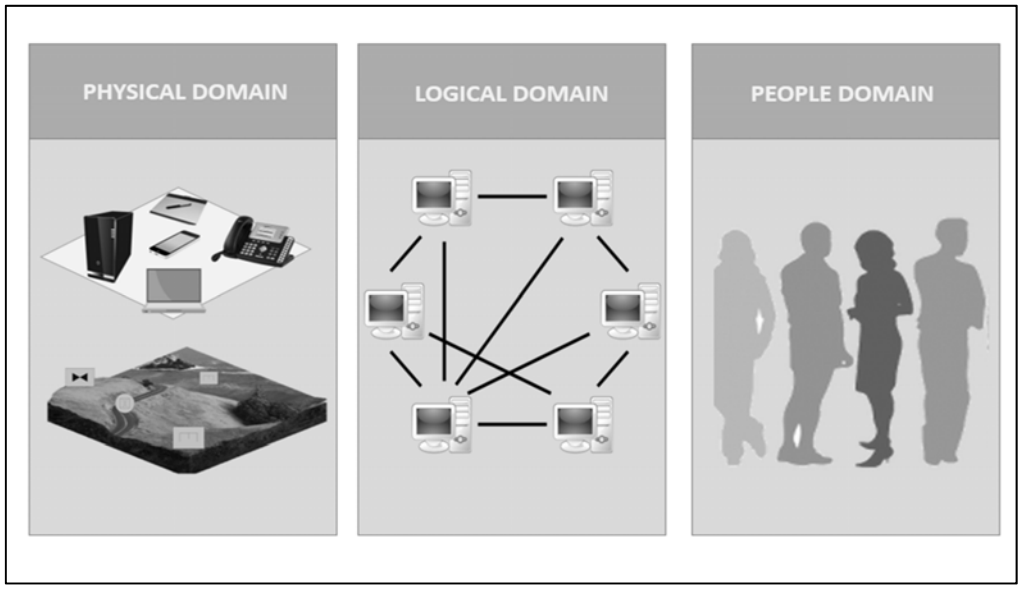


Figure 1.1 – Domains of Cyberspace Operational Environment (US Joint Chief of Staff, 2018)

0104. Like traditional military operations, the CO has a defined OE. The PA cyberspace OE is described as an environment where there is PA cyber presence. It is divided into three domains namely:

a. **Physical Domain.** This refers to the geographical characteristic of the OE like the man-made and natural features. The physical domain includes the location of Army units, assets, and facilities. It also refers to cyber assets like computers, network devices, and gadgets present in the OE.

b. **Logical Domain.** This refers to the network topology and configurations. The logical domain is the relationship and interaction of cyber assets within an information network other than its physical connectivity.

c. **People Domain.** This refers to all persons behind every cyber assets operating within the OE. They are the people who create, move and consume information in and around the PA cyberspace.

Threat Actors

0105. Operating within the OE are actors that launch cyber-attacks against the Army information network infrastructure. These attacks may hinder the accomplishment of the Army's objectives or even bring harm to friendly forces. Threat actors are classified as follows:

a. Nation state actors are those which conduct state sponsored cyber-attacks. They are usually well supported and are motivated by political, economic, technical, or military agendas

b. Cyber criminals are actors who conduct attacks in support of criminal activities. They are usually motivated by money

c. Hacktivists are actors which conduct attacks in support of political, religious and ideological belief. They are neither motivated by money nor power. Albeit, this group believes that what it is doing is noble and for a greater good.

d. Terrorist groups are actors similar to hacktivists except, the motivation for these groups is to inflict harm and sow fear among its targets

e. Thrill seekers, also known as script kiddies, are actors who conduct attacks for the mere fun of it. They are usually motivated by gaining bragging rights to every successful attack

f. Insider threats are actors that can either be intentional or unintentional attackers. Intentional attackers are usually disgruntled employees who wish to do harm or exact revenge on a specific person or to the organization as a whole. Unintentional attackers are unwitting victims of other threat actors, who has somehow gained control over the victim's credentials or workstation, and use them as platforms for attack.

PA Cyberspace Operations Mission

0106. In order to address the challenges in the cyberspace OE, the Army conducts PA Cyberspace Operations (PA-CO). The purpose of PA-CO is to allow Army forces to operate and accomplish its objectives in or through cyberspace.

0107. The PA-CO has three broad objectives:

a. Connect Army forces operating in the physical environment to the PA information network infrastructure, also known as the Philippine Army Network (PANET).

b. Secure connectivity to deny threat actors from compromising mission critical services and information systems.

c. Dominate cyberspace at the OE by denying and disrupting enemy cyberspace capabilities.

0108. These objectives are divided into three distinct missions namely:

a. Information Network Operations (Connect)

b. Defensive Cyberspace Operations (Secure)

c. Active Defense Operations (Dominate)

0109. The following table shows the distinction between missions based on their objectives and activities:

Table 1.1 – Cyberspace Operations Missions, Objectives and Activities

Cyberspace Missions	Objective	Activities
Information Network Operations (INO)	Build, configure, secure, operate, maintain and sustain	• Cyber Operational Preparation of the Environment (C-OPE) • Network Operations Center Monitoring • Network Recovery Operations • Information Systems Development
Defensive Cyberspace Operations (DCO)	Secure and deny	• Security Operations Center (SOC) Monitoring • Computer Security Incident Response (CSIR) • Cyber Risk Assessment (CRA) • Vulnerability Assessment and Penetration Testing (VAPT)
Active Defense Operations (ADO)	Disrupt and deny	• Cyber Intelligence Preparation of the Battlefield • Cyber Order of Battle Buildup • Cyber Counter Action • Social Media Monitoring • Information Operations (IO) • VAPT

0110. **Information Network Operations (INO)**. The purpose of INO is to build the PANET and effect the expedient restoration of connectivity and services after a computer emergency.

0111. **Defensive Cyberspace Operations (DCO)**. The purpose of DCO is to defend PANET from all forms of cyber threats and attacks. It also provides prompt and effective response to cyber incidents.

0112. **Active Defense Operations (ADO).** The purpose of ADO is to exploit the enemy cyber vulnerabilities in order to defeat and deny them the ability to use cyberspace against Army forces.

PA Cyberspace Operational Framework

0113. The PA CO framework represents the global expansion of the PA's OE with cyberspace operations through its three missions, INO, DCO, and ADO. The three missions share space with the traditional PA OE, where they are launched. Once operationalized, these missions expand the PA's physical, logical and people domain, from the traditional OE into the global cyberspace domain. In the global cyberspace domain are the threat actors that are constantly trying to penetrate PANET. Note that insider threats are inside the people domain, as they are part or members of the organization.

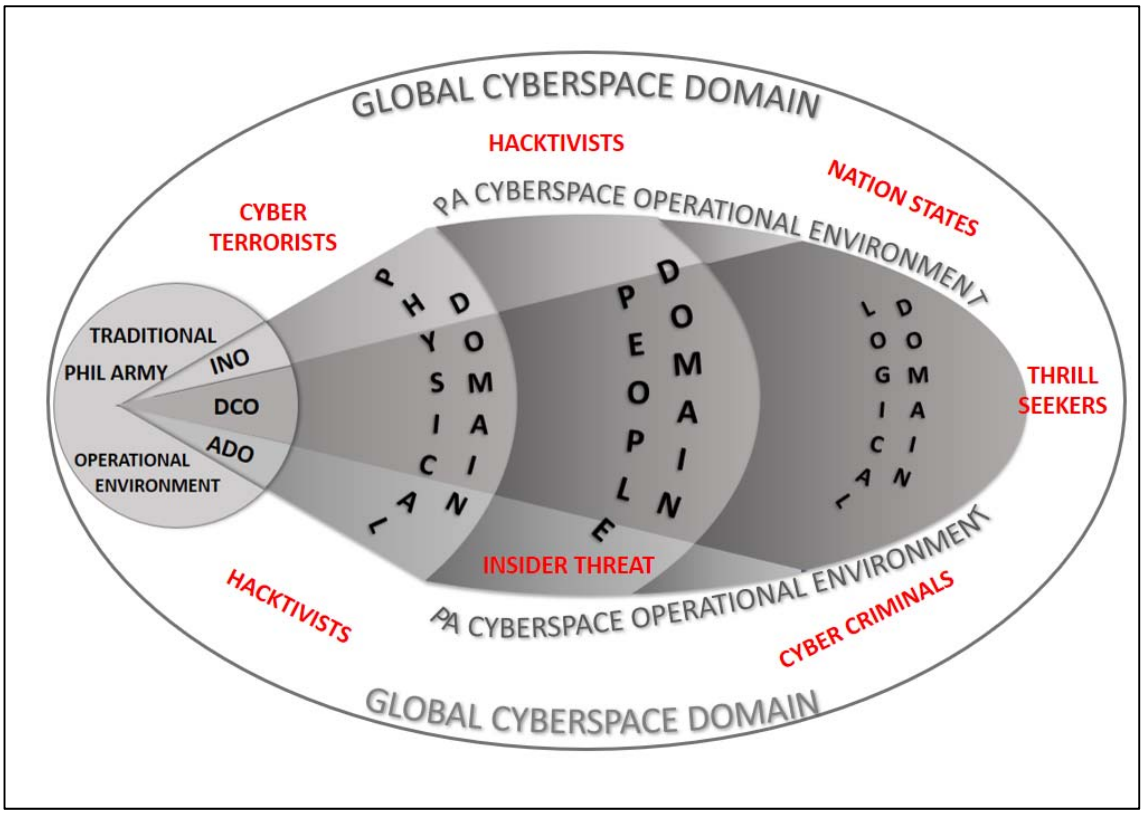


Figure 1.2 - PA Cyberspace Operational Framework

Chapter II
Fundamentals of Cyberspace Operations

Concepts and Strategies

0201. The principles of PA-CO are based on the concepts and strategies found in existing manuals and publications. These documents serve as guide in the formulation of PA-CO's concept of operation, making sure that everything is aligned and compliant.

0202. **Information Security Governance and Risk Management**. This is the strategy embodied in the National Cybersecurity Plan 2022. It is a systematic and methodical practice of ensuring the protection of mission-critical and non-mission critical infostructure (Department of Information and Communication Technology, 2017).

0203. **Joint Cyberspace Operations and Cyberspace Operations in support to AFP Joint Operations**. These concepts are embodied in the AFP Cyberspace Operations Manual. The joint cyberspace operations is the fusion of cyber forces from the different major services (MS), whose task is to address threats and attacks. On the other hand, cyberspace operations in support to AFP Joint Operations are those which directly support the forces engaged in joint operation. In both cases, the AFP Cyber Group (AFPCyG) acts as the unit of primary responsibility (Office of the Assistant Chief of Staff for CEIS, J6, 2018).

0204. **Information Superior Army**. The PA manual describes the multi-dimensional role of Army forces and the future challenges of becoming an information superior Army. Information superiority allows access to accurate and real-time information (Phiilippine Army, 2009).

0205. **Information Superiority in Full Spectrum Operations**. PAM 3-00 introduces the concept of information superiority as part of enabling operations, in support to PA full-spectrum operations. Information superiority can be achieved by using information technology that provides real time information and improves the commander's situational understanding (AC of S for Operations, G3, 2014).

0206. **Cyber Defense**. The Philippine Army Operating Concept (PAOC) included threats to cyberspace as part of the PA's strategic environment. Said threats are addressed using the cyber defense concept. Cyber defense refers to the protection of both command and control, communications, computers, intelligence, surveillance, and reconnaissance (C4ISR) infrastructure and data in cyberspace (Philippine Army, 2015).

0207. **Information Operations (IO).** In the PA Landpower Maneuver Concept (LMC), IO is regarded as the PA's information-communication-technology strategy in achieving information superiority. It provides the PA the ability to operate in the information environment. IO's objective is to protect and defend the land force and information capabilities, infrastructure and systems. Its primary task is the establishment of C4ISTAR which shall provide the Army the capability of a robust

command and control system and intelligence fusion for distributed intelligence operation (Philippine Army, n.d.).

0208. **Information Systems and Cyber Systems**. PAM 6-00 considers these two systems as part of the C4S sub-systems. Information systems are computer-based processes of sourcing, manipulating and storing data, while cyber systems establish the classification, downgrading, declassification, transmission, transportation, and safeguarding of information. (AC of S for C4S, G6, 2017).

Cyberspace Operations Activities

0209. Three principles stood out in the preceding concepts and strategies: continuity of service; defense of cyber systems; and information superiority. They serve as the guiding principles in the planning and execution of the three CO missions (INO, DCO and ADO). Each mission has specific activities integral to their success. On top of this, is a capstone activity which harmonizes the three missions altogether.

Information Network Operations Activities

0210. INO lays the groundwork for building and sustaining a pervasive and resilient information network infrastructure. Activities under this operation includes: Cyber Operational Preparation of the Environment; Network Operations Center Monitoring; Network Recovery Operations; and Information Systems Development.

0211. **Cyber Operational Preparation of the Environment (C-OPE)** is adopted from the US Manual JP 3-13. These are activities conducted to plan and prepare an OE for cyberspace operation. It includes but is not limited to identifying data, system/network configurations, or physical structures connected to or associated with the network or system to include software, ports, and assigned network address ranges or other identifiers (The Vice Chairman of the Joint Chief of Staff, USAF, n.d.).

0212. **Network Operations Center Monitoring** are activities which ensure continues PANET connectivity as well as monitor the health of servers and network devices at the PA Data Center.

0213. **Network Recovery Operations** are activities done during or after the occurrence of computer emergency situation. The main objective of this activity is to restore PANET connectivity and all its services.

0214. **Information Systems Development** are activities which refer to the development, integration and maintenance of Critical and Non-Critical Information Systems which run inside the PA Red Network.

Defensive Cyberspace Operations Activities

0215. DCO commences at the same time with INO or while in the process of building the network. Security and monitoring systems are emplaced during this early phase of operation, where the Army network is most vulnerable. DCO has three main activities:

Security Operation Center Monitoring; Computer Security Incident Response; Cyber Risk Assessment; and Vulnerability Assessment and Penetration Testing.

0216. **Security Operations Center (SOC) Monitoring**. One of the main activities of DCO is the establishment of a SOC, a generic term which describes a platform that provides detection and reaction services during security incidents. It has five modules namely: security event generation, collection, storage, analysis and reaction (Bidou, n.d.).

0217. **Computer Security Incident Response (CSIR)** are activities done during or after security incidents or attacks. The main objective of this activity is to mitigate the effects of the attack or prevent it altogether. A thorough investigation is conducted during the response to determine the extent of compromise and attribution of the attack. The Army is adapting the NIST framework in CSIR.

0218. **Cyber Risk Assessment (CRA)** are activities which test the compliance of the PA Network infrastructure to the Center for Internet Security Controls version 7.1 standards (Center for Internet Security, 2019).

0219. **Vulnerability Assessment and Penetration Testing (VAPT) Activities** are two security testing services that focuses on identifying vulnerabilities in the network, server and network infrastructure (Singh, 2018). While mainly an ADO activity, VAPT is conducted during DCO mission purposely for testing internal information network infrastructure.

Active Defense Operations Activities

0220. While the Army does not condone nor support any form of cyber warfare, ADO activities remain relevant during exercises and primarily for self-defense especially when the civilians and soldiers' lives are at stake. ADO activities include:

a. **Cyber Intelligence Preparation of the Battlefield** is derived from the Intelligence Preparation of the Cyber Environment (IPCE). It is a systematic and continuous process of analyzing; the means and motives of threat actors; our digital environment and the digital environment in which we operate; in order to understand the likely scenarios in which we will face threats, enhancing our operational resiliency (Dartnall, 2018).

b. **Cyber Order of Battle Buildup** is analogous to a kinetic warfare order of battle. It provides an overview of the cyberspace capabilities of the adversary to include groups or individual persons of interests identified as threat actors.

c. **Cyber Counter Action** are activities directed towards attacks that pose as a threat to national security or compromises army personnel during combat missions. It is geared primarily towards the protection of Army units by conducting direct action against said attackers. Cyber counter action also includes the filing of appropriate charges against identified perpetrators.

d. **Social Media Monitoring** are activities which monitor various social media platforms that may be used by identified threat groups as a medium to recruit or support terroristic activities.

e. **Information Operations (IO)** are activities which focus more on the internal and external perception aspects of the PA infostructure. Its purpose is to influence, disrupt, corrupt, or usurp the decision making of the adversaries while protecting its own (US Joint Chief of Staff, 2014).

f. **Vulnerability Assessment and Penetration Testing (VAPT)** during ADO missions are directed towards identified threat actors.

Capstone Activity

0221. **Cyber range** is a capstone activity which tests the three cyberspace operation missions altogether. Cyber ranges are interactive, simulated representations of an organization's local network, system, tools, and applications that are connected to a simulated Internet level environment. They provide a safe, legal environment to gain hands-on cyber skills and a secure environment for product development and security posture testing (National Institute of Standards and Technology, n.d.). This activity may be done internally in the Army, or with other major services, civilian entities, and allied forces.

Roles and Responsibilities

0222. Cyberspace operations at the joint level shall be undertaken by a cyber unit (CU) at the General Headquarters, AFP. Said unit conducts cyberspace operations through planned, coordinated, integrated and synchronized efforts in support of the AFP joint operations and the protection of the AFP's info-structure (Armed Forces of the Philippines, 2018).

0223. At the highest level of command in the PA, a committee chaired by the Chief of Staff, PA, and vice-chaired by the Assistant Chief of Staff for Command, Control, Communications, Computer and Cyber Systems, G6 (AC of S for C4S, G6), PA is formed. Said committee deliberates proposed strategies, policies, plans, programs and projects for PA operations in cyberspace (Philippine Army, 2018).

0224. The AC of S for C4S, G6 at the Headquarters PA (HPA) facilitates the formulation of policies and guidelines pertaining to the conduct of cyberspace operations as well as oversees PA-wide cybersecurity requirements and activities.

0225. The OG6 at the PAMU and subordinate units, oversees the cybersecurity requirements and activities in their respective levels of command.

0226. Below the command and policy level at HPA is a battle staff that serves as the working arm of the PA Cyber Committee (Philippine Army, 2019). This ad-hoc body provides concerted efforts for conceptualization, planning, execution and decisions of the Command on cyber incidents and operations. The functions of said Battle Staff include:

a. Conceptualize and plan for PA-wide cyber operations.

b. Conduct regular review on cyber incidents that occurred in the PA network and provide updates on the future cyber threats that needs to be addressed by the cyber units of the PA.

c. Discuss and formulate techniques, tactics and procedures (TTP) on the conduct of cyber operations.

d. Convene immediately whenever a major cyber incident occurs and provide appropriate recommendations.

0227. The Army Signal Regiment, PA provides general support cyber units, separate and distinct from the traditional Field Signal Battalions, purposely to undertake INO and DCO functions in support of combined arms operation and in defense of the PA info-structure.

0228. Integral to the conduct of cyberspace operations is the creation of the Computer Security Incident Response Teams (CSIRT), deployed across the different echelons of the PA (Network Enterprise Technology Battalion, 2019):

a. **PA CSIRT** is composed of highly trained and highly equipped personnel from the ASR, PA.

b. **PAMU CSIRT** is an ad-hoc team that is activated during security incidents at the operational level. It is composed of trained and equipped personnel from the Field Signal Battalion. PAMU CSIRT also caters to the cybersecurity requirements of the brigades and battalions.

0229. The Army Intelligence Regiment is the primary unit responsible in conducting ADO for intelligence in the PA.

0230. Civil Military Operations Regiment is the primary unit responsible in conducting ADO for IO in the PA.

0231. Information system officer (ISO) and information system non-commissioned officers (ISNCO) are front runners for CO at the different offices and PA units.

0232. Information technology (IT) partners are civilian entities to include technology vendors. They keep the Army abreast with the latest technologies in the market. IT partners with expertise in cybersecurity are also tapped during emergency situations.

Chapter III
Planning and Execution

0301. The PA-CO undergoes the same operational planning process of analyzing the current state of the OE, identifying the threats, defining an end state, and developing lines of operations to achieve the desired state.

State of the Operational Environment

0302. Careful analysis of the present condition of the OE enables the Army to identify the gaps that hinder it from achieving its objectives. The following describes the current state of the cyberspace OE. This is followed by the desired state which answers the gaps identified in the current state.

Current State of the Operational Environment

0303. **Unstable PA network services**. The Army network is dependent on the services of commercial Telecommunication Companies (Telco). During natural and man-made calamities, Telco facilities are often the first to go down. When these facilities go down, so does the PA network. Restoration of services often takes a long time.

0304. **Persistent Cyber Threats present in the PA network**. Cyberattacks are becoming prevalent in the PA network. Threat actors are becoming more and more sophisticated in their methods and approaches that despite cybersecurity efforts Advance Persistent Threats (APT) continue to find its way into the PA network.

0305. **Attackers remain anonymous and unaccountable**. Despite some success in defending the PA network from cyberattacks, perpetrators remain anonymous and unscathed. The lack of direct action against the attackers emboldens them to keep on trying.

Desired State of the Operational Environment

0306. **Pervasive and resilient network**. In this state, PA network is highly available and is able to withstand and quickly recover from computer emergencies brought about by natural and man-made calamities.

0307. **Formidable network security emplaced**. A responsive cybersecurity system, coupled with efficient incident response capabilities, which effectively address any form of cyber-attacks.

0308. **Cyber-attackers neutralized**. Attacker's ability to launch a cyber-attack is neutralized. In addition, the identities of attackers are revealed and appropriate legal actions are filed against them.

Current Threats

0309. At present, nation states, cyber terrorists, thrill seekers and hacktivists are the active threat actors in the PA cyberspace. Other threat actors still operate in the OE and are given ample attention. In addition to threat actors, natural and manmade calamities, which causes destruction of communication facilities, also pose a threat to the PA network infrastructure.

End State

0310. The PA-CO's end state is hinged on the unhampered, timely and secured delivery of critical information network services to Army forces. The PA-CO's desired end state can be stated simply as: *A pervasive and formidable Army Network.*

Phases of Cyberspace Operations

0311. To bridge the gap between the current and the desired state, lines of operations are drawn across four phases of operation. The PA-CO base its phases of operation with the crisis management cycle. Note in the PA-CO, planning and preparation phase takes precedence over mitigation phase. The PA-CO phases of operation are:

a. **Planning and Preparation Phase**. This phase lays the foundation for the overall conduct of CO. Each CO mission requires a planning activity to prepare the OE for CO. It is also in this phase that the Army starts building the network and security systems using the different cyberspace capabilities on hand.

b. **Mitigation Phase**. This phase operationalizes monitoring facilities (i.e. Network Operation Centers and Security Operation Centers) to safeguard the systems built from the previous phase. It is also in this phase that quality assurance activities are conducted to allow proactive discovery of vulnerabilities, so that they may be addressed early in the operation.

c. **Response Phase**. This phase occurs after a computer emergency incident has occurred. During this phase, the CSIRT are called to action. The type of CSIRT (HPA, PAMU or Brigade) that will be called will depend on where the attack was detected and the severity of the attack.

d. **Recovery Phase**. This phase happens after threats have already been addressed. The main objective of this phase is to restore connectivity and services affected by the incident.

Cyberspace Operational Design

0312. From the defined phases of operation, lines of operations are drawn. The lines of operations are derived from the cyberspace operations missions: INO, DCO and ADO. Activities within each mission comprise the operational approach, bridging the current state with the desired state.

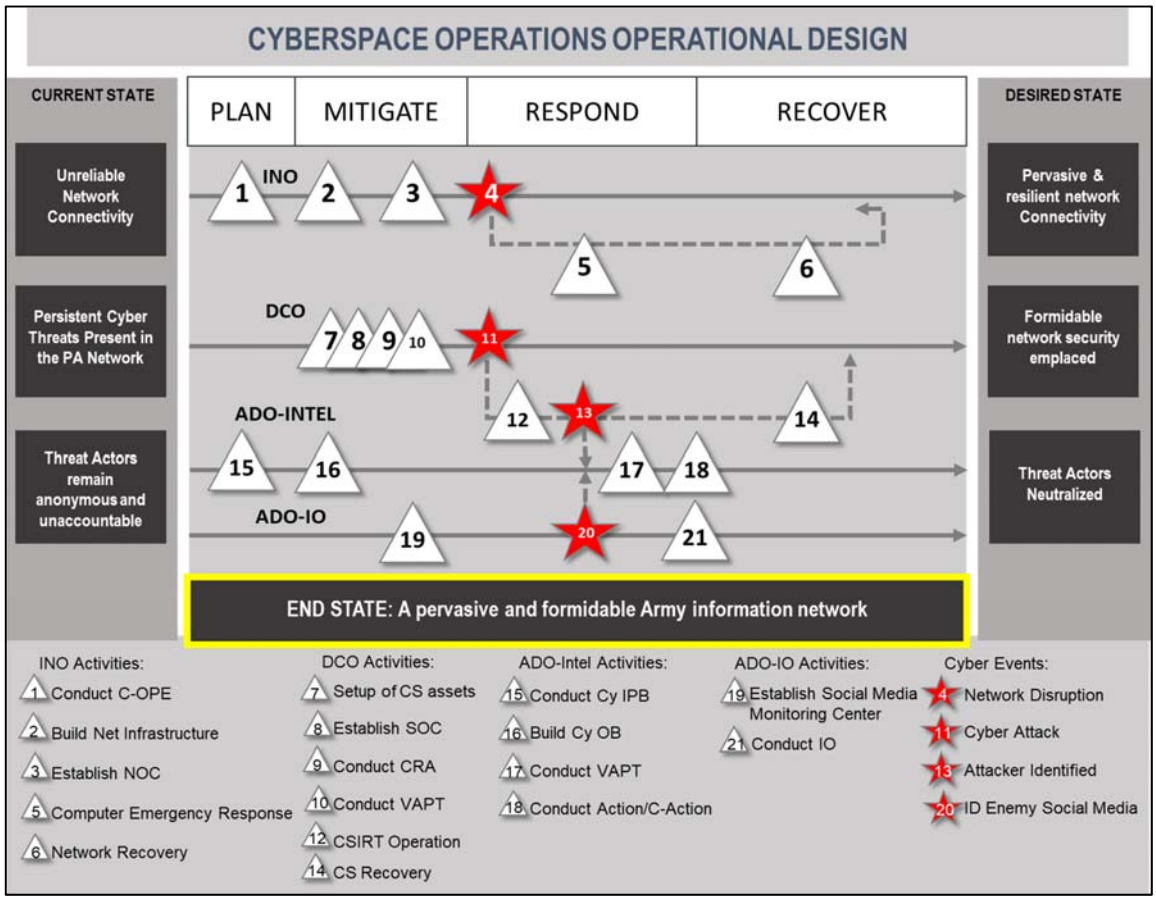


Figure 3.1 – Cyberspace Operational Design

0313. The operational design (OD) serves as a template for planning and execution of cyberspace operations in support of PA operations. Implementation of the various operational approaches may vary depending on the mission and available cyberspace capabilities yet the objectives of the CO remain the same.

PA Cyberspace Operations by Echelon

0314. PA-CO is conducted across the different echelons of the PA. Units responsible per echelon has specific tasks and responsibilities. The following figure illustrates CO in the different echelons of the Army.

Strategic	HPA	CyBn/NETBn/AIR/CMOR (PANET, PA Data Center)
		IS Officers/IS NCOs (Offices)
Operational	PAMU	IT Spt Pltn, FSBn/MIBn/CMOBn (PANET, PAMU Data Center)
		IS Officers/IS NCOs (OG6)
Tactical	Brigade	IS Sqd, BSSC/MICO/CMO Coy (Deployable Bde TOC System)

Figure 3.2 – PA Cyberspace Operations by Echelon

0315. CSIRT by echelon requires minimum sets of skills in order to perform the response. The following table shows the different CSIRT skills required by echelon.

Table 3.1 – CSIRT Capabilities by Echelon

CSIRT	PA	PAMU	Brigade
Team leader/Incident Manager	✓	✓	✓
Incident Responder		✓	✓
Photographer/Evidence Manager	✓	✓	
Forensic Investigator	✓		

PA Cyberspace Operations in Joint Operations

0316. The PA-CO's responsibility is the PA information network infrastructure. In joint operations however, it is also expected to support AFPCyG's joint cyberspace operations and cyberspace operation in support of AFP joint operations.

0317. In joint cyberspace operations, incident response teams of the different major services shall be integrated under the AFPCyG in order to coordinate efforts in addressing a threat or attack. While in cyberspace operations in support of AFP Joint Operations, the task of cyber units shall be dependent on the mission and intent given by the joint force commander. In addition, in this type of operation, the joint force commander may require specific major service cyber unit for support.

PA Cyberspace Operations and the PA C4ISTAR

0318. PAM 6-00 defines C4ISTAR systems as the integration of command, control, communications, computer, intelligence, surveillance, target acquisition and reconnaissance functions to arm the commander at all levels of command with accurate, timely and correct information directly from the battle ground (AC of S for C4S, G6, 2017).

0319. An effective C4ISTAR requires the efficient and timely exchange of data or information across a web of systems using mostly internet protocol (IP), microwave, and radio frequency technology. Using these technologies makes C4ISTAR systems susceptible to cyberattacks. PA-CO through its different missions, allows the secured and unhampered exchange of information across the different C4ISTAR platforms.

Chapter IV
Sustaining Cyberspace Operations

Personnel Management

Recruitment

0401. Officers and enlisted personnel (EP) may be recruited from the regular PA recruitment process. However, due to the highly technical nature of the CO, the PA encourages alternative methods, like special recruitment, in order to entice highly skilled and highly trained individuals to enter the ranks of the PA cyber workforce. Recruitment is not limited to officers and EP; it is also encouraged to strengthen the cyber workforce with the civilian employees.

Career

0402. Officers and enlisted personnel shall have an initial AFP Occupational Specialty (AFPOS) of Infantry prior to taking a basic career course. After their basic, they shall be inducted into their chosen AFPOS. The cyber workforce for INO and DCO may only be drawn from the Signal Corps (SC) and Military Intelligence (MI). While the cyber workforce for ADO shall be drawn from MI and Infantry AFPOS.

0403. Signal enlisted personnel may also take on a Military Occupational Specialty (MOS), depending on their skill. The following table shows the list of available MOS for CO.

Table 4.1 - Subdivision Signal Corps AFPOS (AC of S for C4S, G6, 2017)

Signal Areas	Military Occupation Specialties (MOS)	Code
Information Systems	Information Technology Technician	22Q
	Information Technology Specialist	22R
	Network Systems Operator/Maintainer	22S
	System/Network Administrator	22T
	Software Developer/Programmer	22U
	Storage and Data Specialist	22V
Cyber Systems	Cyber Operations Specialist	22W
	Cyber Network Defender	22X
	Network Security Operator/Maintainer	22Y
	Computer Incident Responder	22Z

Training

0404. Aside from the career courses, officers and personnel shall undergo specialized courses and trainings in a laddered fashion taking into consideration their ranks and positions. These courses shall be offered at the Signal School (TSS) PA, as part of the In-service Training. Army personnel may also avail of these trainings in other accredited institutions. The following figure illustrates the progression of trainings across an individual's career.

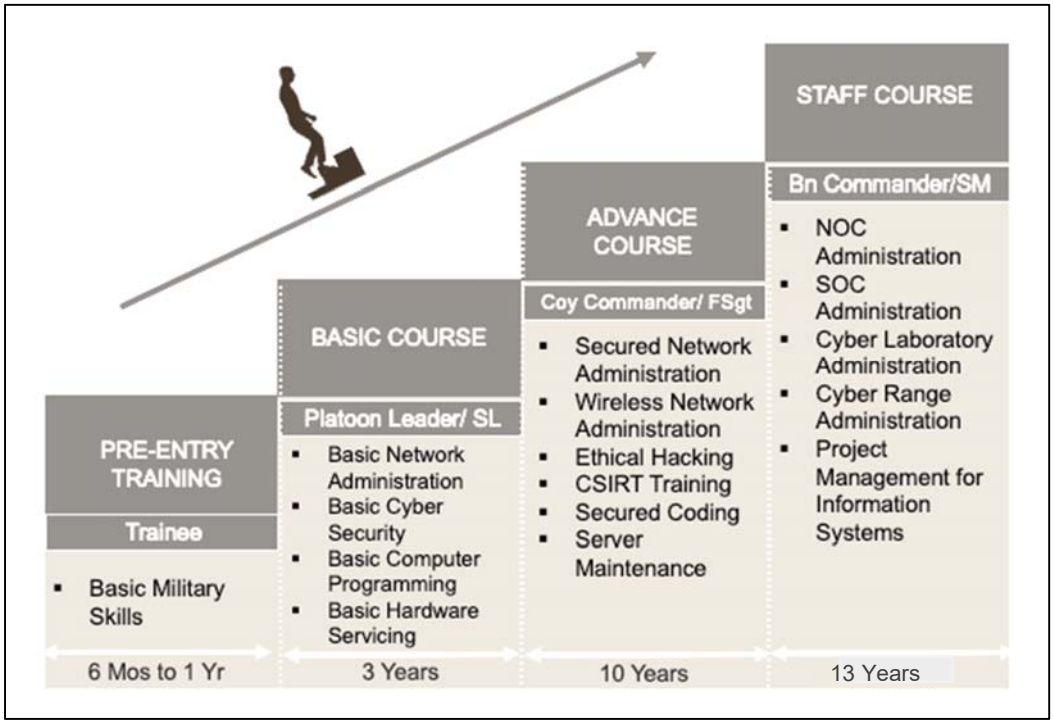


Figure 4.1 – Laddered Career and Skills Diagram for Cyber Workforce

Cyberspace Operations Mission Essential Equipment

0405. CO is a technology-heavy operation. Its success relies heavily on equipment that are essential to the accomplishment of the PA-CO objectives. These mission essential equipment includes, but are not limited to, the following:

a. **Security Information and Event Manager (SIEM)**. A software solution that aggregates and analyzes activity from many different resources across your entire IT infrastructure (Petters, 019).

b. **Vulnerability Assessment Tools**. An assessment solution deployed for identifying the vulnerabilities, configuration issues, and malware that attackers use to penetrate one's own network, or customer's network.

c. **Unified Threat Management (UTM) Firewall**. A networking device or software program which incorporates anti-malware, content filter, firewall, intrusion detection, and spam protection into a single package (Computer Hope, 2018).

d. **IT Monitoring and Management System**. A tool used to quickly detect, diagnose, and resolve performance issues before outages occur and delivers real-time views and dashboards that enable system administrators and network engineers to visually track and monitor network performance at a glance.

e. **Network Access Control (NAC)**. Network access control, or NAC, solutions support network visibility and access management through policy enforcement on devices and users of corporate networks. (Cisco, n.d.)

f. **Active Directory Server**. Active Directory Domain Services (AD DS) is a server role in Active Directory that allows admins to manage and store information about resources from a network, as well as application data, in a distributed database. AD DS can also help admins manage a network's elements (computers and end users) and reorder them into a custom hierarchy.

g. **Anti-spam Firewall**. An all-in-one control of encryption, data loss prevention (DLP), anti-spam, and advanced threat protection.

h. **End Point Security**. A comprehensive endpoint security product with antivirus, antispyware, host intrusion prevention, firewalling, application control, device control, and network access control.

Equipment Life Sustainment Program

0406. The maintenance of mandatory subscriptions for both hardware and software is very crucial to the operation of the PA information network infrastructure components. This must be part of the PA's annual budget as recurring expenses. These mandatories include the following:

a. Firewall Subscription

b. NOC Monitoring System Subscription

c. SOC Monitoring System Subscription

d. End Point Security Subscription

0407. In addition, maintenance and repair of IT hardware should be taken into consideration. At least 30% of the actual cost per hardware should be programmed for repair and maintenance.

0408. Replacement of hardware and software should be done at least every after 5 years from the time the items are purchased.

Glossary

Active Defense Operations
Exploit the enemy cyber vulnerabilities in order to defeat and deny them the ability to use cyberspace against Army forces.

Applications
These are software programs that run on the computer.

Advanced Persistent Threat
Prolonged and targeted cyberattack in which an intruder gains access to a network and remains undetected for an extended period of time (Tech Target, 2018).

Connectivity
Means of connecting one computer/network to another computer/network. It may be through the use of cable and wires, or through wireless means.

Countermeasure
Actions taken to counteract dangers or threats in the cyberspace.

Cyber Criminals
Threat actors which conduct attacks in support of criminal activities.

Cyber Defense
Protection of both C4ISTAR infrastructure and data in the cyberspace.

Cyber Laboratory
A facility for testing and studying cyber threats and/or attacks.

Cyber Risk Assessment
Activities which tests the compliance of the PA Network infrastructure to the Center for Internet Security Controls.

Cyber Systems
Any combination of facilities, equipment, personnel, procedures, and communication to provide cyber services (IT Law, n.d.).

Cyberspace
A global domain within the information environment consisting of the interdependent network of information technology infrastructure and residing data, including the internet, telecom networks, computer systems and embedded processors and controllers (Theohary, 2018)

Cyberspace Assets
Useful or valuable things or persons in the cyberspace.

Cyberspace Capabilities
A device, computer program, or technique, designed to create an effect in or through cyberspace (NIST, n.d.).

Cyberspace Operations
Is the employment of cyberspace capabilities, where the primary purpose is to achieve objectives in or through cyberspace (US Joint Chief of Staff, 2018).

Data Center
A facility that houses the computers, servers, storage, and network systems.

Defensive Cyberspace Operations
Defend PANET from all forms of cyber threats and attacks. It also provides a prompt and effective response to cyber incidents.

Domain
An area in which one has control over.

Exploitation
An action to neutralize cyber-attacks by using adversaries' own vulnerabilities and weaknesses in defending our own network.

Hacktivists
Threat actors which conducts attacks in support of political, religious and ideological belief.

Incident Response
Actions by an organization during or after a cyber-attack or incident takes place.

Information Assurance
Is the practice of making sure that the information that runs through the network remains secure and protected.

Information Network Operations
Groundwork for building and sustaining a pervasive and resilient information network infrastructure.

Information Operations
Information-communication-technology strategy in achieving information superiority.

Information Security
The processes and tools designed and deployed to protect sensitive business information from modification, disruption, destruction, and inspection (Cisco, n.d.).

Information Superiority
Uninterrupted access to real-time information.

Information System
The collection of technical and human resources that provide the storage, computing, distribution, and communication for the information required by all or some part of an enterprise. (Rouse, 2015)

Information Technology
The technology involving the development, maintenance, and use of computer systems, software, and networks for the processing and distribution of data (Miriam Webster, n.d.).

Infostructure
The design of an information resource in the cyberspace.

Infrastructure
Physical structures, buildings, or facilities.

Insider Threat
A member of the organization who intentionally or unintentionally does harm to the network or systems.

Internet
Worldwide series of interconnected computers providing users with various data and information.

Intranet
A private network of interconnected computers within an organization for sharing of digital resources and information.

Logical Domain
Refers to the network topology and configurations.

Nation State
Threat actors are which conducts state sponsored cyber-attacks.

Network
A set of interconnected computers for communication or sharing of resources.

Network Administrator
Person in-charge of monitoring and maintaining the computer network of the office or unit.

Network Security
Is the set of rules that maintains the confidentiality, integrity, and accessibility of information running through the network.

Philippine Army Network
The intranet of the Philippine Army that provides various automated services.

Penetration Testing
Testing a computer network, website, web services, and other systems for vulnerabilities that may be exploited by attackers.

People Domain
Refers to all persona behind every cyber assets operating within the OE.

Physical Domain
Geographical characteristic of the OE i.e. area and natural features.

Platform
Basic hardware or software to initiate processes or applications to run.

Risk Management
Technologies and policies that address threats or vulnerabilities in networks or computer programs and data, which includes, but not limited to, information security, information assurance, and hardware and software assurance.

Script Kiddies
Same as Thrill Seekers.

Security Appliances
An appliance that provides security of the network from unwanted network traffic.

Security Operations Center
A facility that provides IT security to the network of the organization to prevent cyber threats and respond to any cyber incident.

Social Media
Online applications that enables users to create and share profiles, contents, and other information or to participate in social networking or online communication.

System Administrator
Person in-charge of monitoring and maintaining of a certain service or system being offered by the network.

Terrorist Groups
Threat actors similar to hacktivists except, the motivation for this group is to inflict harm and sow fear among its targets.

Threat Actors
Are those that launches cyber-attacks against the Army information network infrastructure.

Threat Analysis
Process in which the knowledge of internal and external information vulnerabilities pertinent to a particular organization is matched against real-world cyber-attacks (Infosec, 2019).

Threat Awareness
Identification and analysis of potential and existing cyber threats within the network and the organization thereby.

Thrill Seekers
Threat actors which conducts attacks for the mere fun of it.

Tools
Hardware or software that ease the conduct of cyberspace operation.

Vulnerability Assessment
The process of identifying, quantifying, and prioritizing risks and vulnerabilities found in computer networks, applications, services, hardware, and software.

Workstation
Computer terminal used by personnel in the office/unit.

Abbreviation and Acronym List

AC of S	Assistant Chief of Staff
ADO	Active Defense Operations
AFP	Armed Forces of the Philippines
AFPCyG	Armed Forces of the Philippines Cyber Group
AFPOS	Armed Forces of the Philippines Occupational Specialty
APT	Advanced Persistent Threats
ASR	Army Signal Regiment
BSSC	Brigade Support Signal Company
C4ISR	Command and Control, Communications, Computers, Intelligence, Surveillance, and Reconnaissance.
C4ISTAR	Command and Control, Communications, Computers, Intelligence, Surveillance, Target Acquisition, and Reconnaissance.
CEIS	Communications, Electronics, and Information Systems
CO	Cyberspace Operations
C-OPE	Cyber Operational Preparation of the Environment
CRA	Cyber Risk Assessment
CSIR	Computer Security Incident Response
CSIRT	Computer Security Incident Response Team
Cy Coy	Cyber Company
DC of S	Deputy Chief of Staff
DCO	Defensive Cyberspace Operations
DICT	Department of Information and Communication Technology
EP	Enlisted Personnel
FSBn	Field Signal Battalion
GHQ	General Headquarters
INO	Information Network Operations
IO	Information Operations
IP	Internet Protocol
IPB	Intelligence Preparation of the Battlefield
IPCE	Intelligence Preparation of the Cyber Environment
IS Sqd	Information System Squad
ISNCO	Information System Non-Commissioned Officer
ISO	Information System Officer
IT	Information Technology
IT Spt Pltn	Information Technology Support Platoon
MI	Military Intelligence
MOS	Military Occupational Specialty
MS	Major Services

NETBn	Network Enterprise Technology Battalion
NIST	National Institute of Standards and Technology
NOC	Network Operations Center
OD	Operational Design
OE	Operational Environment
PA	Philippine Army
PA-CO	Philippine Army Cyberspace Operations
PACOBS	Philippine Army Cyber Operations Battle Staff
PAM	Philippine Army Manual
PAMU	Philippine Army Major Unit
PANET	Philippine Army Network
PAOC	Philippine Army Operating Concept
SC	Signal Corps
SOC	Security Operations Center
Telco	Telecommunications Companies
VAPT	Vulnerability Assessment and Penetration Testing

References

AC of S for C4S, G6. (2017, December 11). Philippine Army Manual 6-00 (2nd Ed). Taguig City.

AC of S for Operations, G3. (2014, May 06). Philippine Army Manual 3-00. Taguig City.

Armed Forces of the Philippines. (2013, August 30). Letter Directive Number 28. *Cyberspace as AFP Domain of Operations*. Quezon City: Armed Forces of the Philippines.

Armed Forces of the Philippines. (2018, May 24). General Orders Number 662. *Activation*. Quezon City: Armed Forces of the Philippines.

Bidou, R. (n.d.). *Security Operation Center Concepts and Implementation.* Retrieved from http://www.iv2-technologies.com/SOCConceptAndImplementation.pdf

Center for Internet Security. (2019, April 1). CIS Controls v 7.1. Retrieved from https://learn.cisecurity.org/cis-controls-download

Cisco. (n.d.). *What is Information Security*. Retrieved from Cisco: https://www.cisco.com/c/en/us/products/security/what-is-information-security-infosec.html

Cisco. (n.d.). *What is Network Access Control?* Retrieved from https://www.cisco.com/c/en/us/products/security/what-is-network-access-control-nac.html

Computer Hope. (2018, November 11). *UTM*. Retrieved from Computer Hope: https://www.computerhope.com/jargon/u/utm.htm

Dartnall, R. (2018). *Intelligence Preparation of the Cyber Environment.* Retrieved from SANS: https://www.sans.org/cyber-security-summit/archives/file/summit-archive-1517245731.pdf

Department of Information and Communication Technology. (2017). *National Cybersecurity Plan 2022.* Metro Manila: Department of Information and Communication Technology.

Infosec. (2019, February 8). *Cyber Threat Analysis (Update 2019)*. Retrieved from Infosec: https://resources.infosecinstitute.com/cyber-threat-analysis/#gref

IT Law. (n.d.). *Cyber System*. Retrieved from IT LAW WIKI: https://itlaw.wikia.org/wiki/Cyber_system

Miriam Webster. (n.d.). *Information Technology*. Retrieved from Miriam Webster: https://www.merriam-webster.com/dictionary/information%20technology

Morgan, J., & DiResta, R. (2018, July 10). *Information Operations are a Cybersecurity Problem: Toward a New Srategic Paradigm to Combat Disinformation*. Retrieved from Just Security: https://www.justsecurity.org/59152/information-operations-cybersecurity-problem-strategic-paradigm-combat-disinformation/

National Institute of Standards and Technology. (n.d.). *Cyber Ranges.* Retrieved from https://www.nist.gov/sites/default/files/documents/2018/02/13/cyber_ranges.pdf

Network Enterprise Technology Battalion. (2019). Cyberspace Operations Presentation. Taguig City: Philippine Army.

NIST. (n.d.). *Computer Security Research Center: Cyberspace Capability*. Retrieved from NIST: https://csrc.nist.gov/glossary/term/cyberspace-capability

Office of the Assistant Chief of Staff for CEIS, J6. (2018). *Cyberspace Operations.* Quezon City: AFP.

Payoree, J., & McKenny, C. (2018). Building a Security Operation Center - for little or no money. Retrieved from https://www.defcon.org/images/defcon-18/dc-18-presentations/Pyorre/DEFCON-18-Pyorre-Building-Security-Operations-Center.pdf

Petters, J. (019, July 4). *What is SIEM? A Beginner's Guide*. Retrieved from Varonis: https://www.varonis.com/blog/what-is-siem/

Philippine Army. (2009, October 16). Philippine Army Manual 1. Taguig City.

Philippine Army. (2015). Philippine Army Operating Concept. Philippine Army.

Philippine Army. (2018, November 9). Letter Orders Number 552. *Termination/Detail*. Philippine Army.

Philippine Army. (2019, May 26). Creation of PA Cyber Operations Battle Staff. Taguig City: Philippine Army.

Philippine Army. (n.d.). Landpower Maneuver Concept. Philippine Army.

Singh, M. (2018, March 9). *What is VAPT vulnerability assessment and penetration testing?* Retrieved from Quora: https://www.quora.com/What-is-VAPT-vulnerability-assessment-and-penetration-testing

Tech Target. (2018, July). *Advanced Persistent Threat (APT)*. Retrieved from TechTarget: https://searchsecurity.techtarget.com/definition/advanced-persistent-threat-APT

Tech Target. (n.d.). *vulnerability assessment (vulnerability analysis)*. Retrieved from TechTarget: https://searchsecurity.techtarget.com/definition/vulnerability-assessment-vulnerability-analysis

The Vice Chairman of the Joint Chief of Staff, USAF. (n.d.). Cyberspace Operations Lexicon. Retrieved from http://www.nsci-va.org/CyberReferenceLib/2010-11-joint%20Terminology%20for%20Cyberspace%20Operations.pdf

Theohary, C. A. (2018, December 18). Defense Primer: Cyberspace Operation. Retrieved from https://fas.org/sgp/crs/natsec/IF10537.pdf

US Joint Chief of Staff. (2014). *Information Operation.* US Armed Forces. Retrieved from https://www.jcs.mil/Portals/36/Documents/Doctrine/pubs/jp3_13.pdf

US Joint Chief of Staff. (2018). *Cyberspace Operations.* United States Armed Forces.